

Practice Midterm 1

CSCI 160: Computer Architecture 1

Instructions. Show work to justify each answer. Partial credit will be awarded.

Part I: Multiple Choice

Circle the best answer.

1. Which statement best describes a digital system? [2 pts]
 - ☐ A. A system that stores only decimal numbers.
 - ☐ B. A system that manipulates discrete information, usually represented internally with 0s and 1s.
 - ☐ C. A system that cannot be programmed.
 - ☐ D. A system that must use hexadecimal instead of binary.
2. A byte contains 8 bits. How many different bit patterns can one byte represent? [2 pts]
 - ☐ A. 8
 - ☐ B. 16
 - ☐ C. 128
 - ☐ D. 256
3. In Boolean algebra, which statement is true? [2 pts]
 - ☐ A. The expression $x + yz = (x + y)(x + z)$ is a valid Boolean identity.
 - ☐ B. The symbol $+$ always means ordinary decimal addition.
 - ☐ C. Boolean algebra has subtraction and division as basic operations.
 - ☐ D. The complement of x is written $2 - x$.
4. Which procedure correctly forms the 8-bit signed 2's-complement representation of a negative number? [2 pts]
 - ☐ A. Change only the leftmost bit from 0 to 1.
 - ☐ B. Invert every bit of the positive version, then add 1.
 - ☐ C. Add 0110 to each group of four bits.
 - ☐ D. Convert the number to hexadecimal and keep the same digits.
5. The XOR operation $x \oplus y$ is 1 exactly when: [2 pts]
 - ☐ A. x and y are both 0.
 - ☐ B. x and y are both 1.
 - ☐ C. x and y have different values.
 - ☐ D. at least one of x and y is 1.

Part II: Show Work

6. **Number-base conversions.** Convert each number as requested. Show grouping, place values, division remainders, or another clear method. [16 pts]

(a) Convert $(101101.011)_2$ to decimal. [5 pts]

(b) Convert $(684)_{10}$ to hexadecimal and then to binary. [6 pts]

(c) Convert $(5B3.8)_{16}$ to binary and octal. [5 pts]

7. **Binary arithmetic and complements.** Show the arithmetic in binary. [15 pts]

(a) Add: $(1011011)_2 + (110111)_2$. [4 pts]

(b) Multiply: $(1011)_2 \times (110)_2$. [4 pts]

- (c) Compute $01011010_2 - 00110111_2$ using the 8-bit 2's-complement method. Clearly show the 2's complement of the subtrahend and the final 8-bit result. [7 pts]

8. **Signed representations and BCD.** Show the steps. [14 pts]

- (a) Convert +23 and +58 to 8-bit signed 2's-complement representation. Then compute $(+23) - (+58)$ using 8-bit signed 2's-complement addition, and convert the result back to decimal. [6 pts]

- (b) Interpret 11111000_2 as an 8-bit signed 2's-complement number. [4 pts]

- (c) Add $145 + 237$ in BCD. Show the BCD addition and final BCD result. [4 pts]

9. **Truth table, minterms, and maxterms.** Let

$$F(A, B, C) = A'B + AC' + B'C.$$

Complete the truth table by first filling in the three expression-part columns. Then write F as both a sum of minterms and a product of maxterms, using expanded notation and shorthand notation. [15 pts]

A	B	C	$A'B$	AC'	$B'C$	F
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

10. **Boolean simplification.** The postulates and theorems below may be used for this problem. Simplify each expression to a minimum number of literals. State the Boolean law or theorem used at each major step. [12 pts]

Postulate	OR / + form	AND / · form
Identity	$x + 0 = x$	$x \cdot 1 = x$
Commutative	$x + y = y + x$	$xy = yx$
Distributive	$x(y + z) = xy + xz$	$x + yz = (x + y)(x + z)$
Complement	$x + x' = 1$	$xx' = 0$

Theorem	Statement
1	$x + x = x, \quad x \cdot x = x$
2	$x + 1 = 1, \quad x \cdot 0 = 0$
Involution	$(x')' = x$
Associative	$x + (y + z) = (x + y) + z, \quad x(yz) = (xy)z$
DeMorgan	$(x + y)' = x'y', \quad (xy)' = x' + y'$
Absorption	$x + xy = x, \quad x(x + y) = x$

(a) $F = A'BC + A'BC' + AB'C + ABC.$ [4 pts]

(b) $G = (x + y)(x + y')(x' + z).$ [4 pts]

(c) $H = wx'y + wx'y' + wxy + w'xy.$ [4 pts]

11. **Boolean identity proof.** Determine whether the following Boolean equation is true or false. Justify your answer using the postulates and theorems of Boolean algebra. Do not use a truth table. [8 pts]

$$AB + A'C + BC = AB + A'C.$$

12. **Logic gates and bitwise operations.** [10 pts]

- (a) Draw a gate-level logic diagram for

$$Y = (A + B')(C \oplus D) + A'C.$$

Use only NOT, AND, OR, and XOR gates. Label all inputs and the output Y . Do not simplify the expression before drawing. [6 pts]

- (b) Let $P = 10101100$ and $Q = 01110110$. Compute $P \oplus Q$, P NAND Q , and P NOR Q bit by bit. [4 pts]